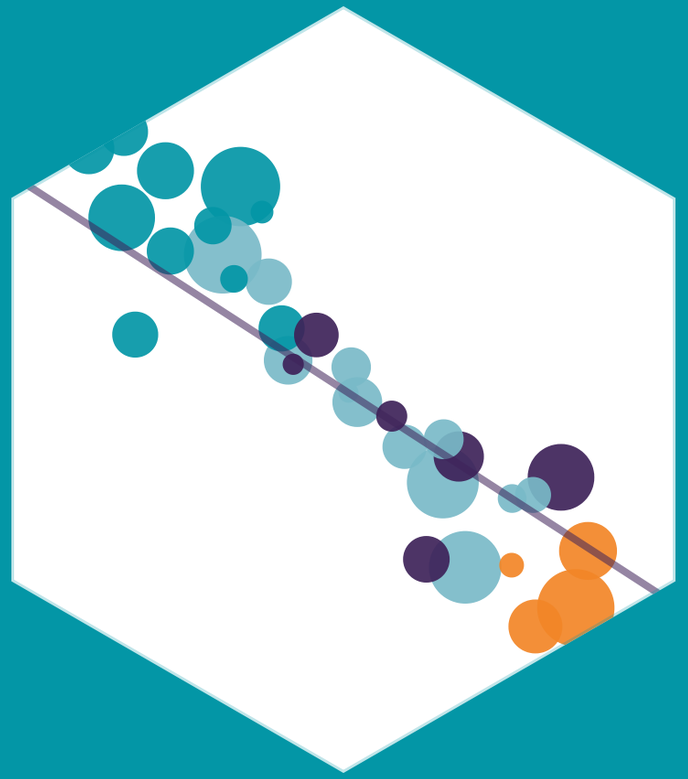
ECONOMETRICS

TA SESSION 7

Barcelona School of Economics
Lucia Sauer
lucia.sauer@bse.eu



Planification — 2° Part

TA6 — RCTs ✓

TA7 — Matching ← today

TA8 — Instrumental Variables

📅 Hand in Assignment 6 by **18th Nov**

TA9 — Regression Discontinuity

📅 Hand in Assignment 7 by **27th Nov**

TA10 — Difference-in-Differences

📅 Hand in Assignment 8 by **2nd Dec**

Today

1. **Observational data** — was the treatment randomly assigned?
2. **Matching on covariates** — nearest neighbour, exact matching
3. **The propensity score** — common support, balancing property, NN / radius / kernel
4. **Back to job training** — which matching method wins, and did it recover the experiment?

💻 Stata: **TA7.do** (data `nsw_cps.dta`) · 📓 Python version: **TA7.ipynb**

1 · Observational data

Did a job-training program raise earnings?

National Supported Work (NSW): a 1970s US program that gave disadvantaged workers (ex-offenders, ex-addicts, long-term unemployed) 9–18 months of subsidized work experience. We want its effect on **earnings in 1978**.

185
Participants
D = 1 · mean 1978 earnings $6,349

15,992
Non-participants
US adults from a survey (CPS) · $14,847

Naive comparison: $\bar{Y}_1 - \bar{Y}_0 = 6{,}349 - 14{,}847 = -8{,}498$ dollars. Did training really **cut** earnings by $8,498? Or are participants and non-participants simply **different people**?

> This is **observational data**: nobody flipped a coin. People ended up in the program (or not) for reasons we don't control. First question, as in TA6: **do the two groups look alike before treatment?**

Was the treatment randomly assigned? A balance check

For each pre-treatment covariate X_k: $H_0 : E[X_k \mid D = 1] = E[X_k \mid D = 0]$, Welch t-test (as in TA6). If D were random, we would reject only ~5% of the time.

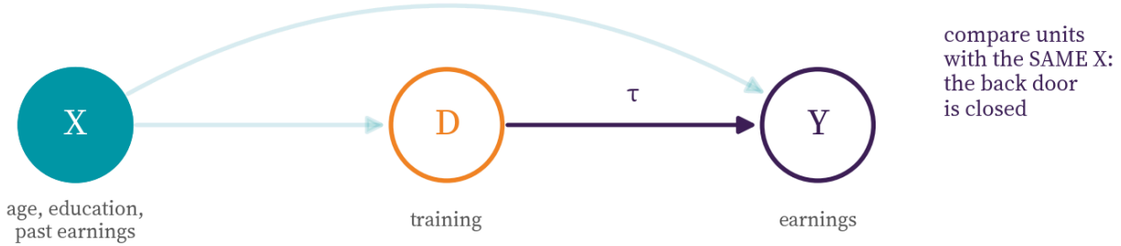
Covariate	Participants	Non-participants	Difference	p-value
Age	25.82	33.23	-7.41	**< 0.001**
Years of education	10.35	12.03	-1.68	**< 0.001**
Black	0.84	0.07	0.77	**< 0.001**
Hispanic	0.06	0.07	-0.01	0.475
Married	0.19	0.71	-0.52	**< 0.001**
No high-school degree	0.71	0.30	0.41	**< 0.001**
Earnings 1974 ($)	2,096	14,017	-11,921	**< 0.001**
Earnings 1975 ($)	1,532	13,651	-12,119	**< 0.001**
Unemployed 1974	0.71	0.12	0.59	**< 0.001**
Unemployed 1975	0.60	0.11	0.49	**< 0.001**

9 of 10 p-values are below 0.05 ⇒ the groups are **not** comparable: clear signs of **non-random** assignment. Participants are younger, less educated, mostly Black, rarely married, and earned about 1/7 of what non-participants earned in 1974–75. And these same variables also determine 1978 earnings ⇒ they are **confounders**.

Without a coin flip, who gets treated is not random

In TA6 randomization made $E[Y \mid D=1] - E[Y \mid D=0] = \text{ATT}$. Without it:

$$\underbrace{E[Y \mid D=1] - E[Y \mid D=0]}_{\text{naive comparison}} = \underbrace{E[Y(1) - Y(0) \mid D=1]}_{\text{ATT}} + \underbrace{E[Y(0) \mid D=1] - E[Y(0) \mid D=0]}_{\text{selection bias}}$$



Idea of matching: compare treated and untreated units **that look the same in X**. Within such a group, who got treated is "as good as random".

Two assumptions

1 · Conditional independence (CIA)

$$\big(Y(1), Y(0)\big) \perp\!\!\!\perp D \mid X$$

Once we compare units with the same X, treatment is as good as random. **Untestable**: we must observe **every** confounder ("selection on observables").

2 · Overlap (common support)

$$0 < P(D = 1 \mid X) < 1$$

For every X there are both treated **and** untreated units: someone to compare with. **Testable**: we look at the data.

For the ATT we only need to impute $E[Y(0) \mid D = 1]$:

$$\text{ATT} \;=\; E[Y \mid D = 1] \;-\; E\Big[\; \underbrace{E[Y \mid X, D = 0]}_{\text{untreated with the same } X} \;\Big|\; D = 1\Big]$$

! Matching is **not** a design like an RCT: it is only as good as the X's you observe. It cannot fix confounding by **unobserved** variables.

2 · Matching

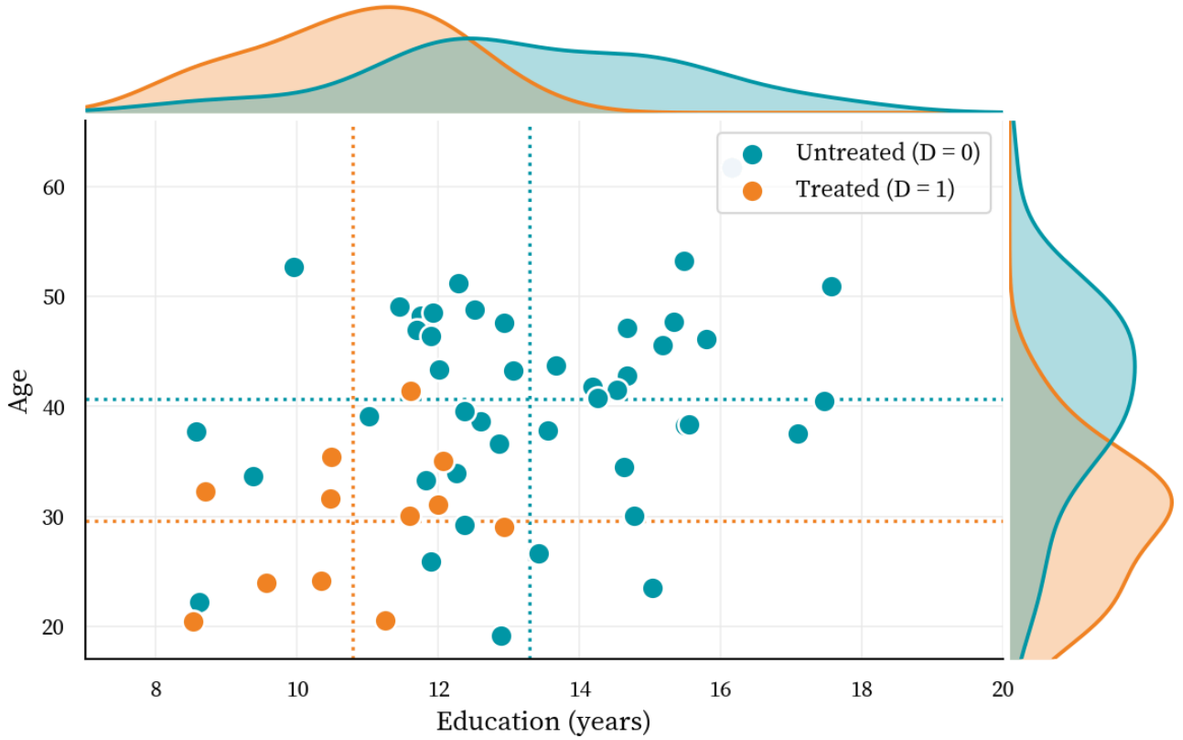
Matching = imputing the missing counterfactual

A toy version of our question (Y = earnings, thousand \$). For each **treated** unit, find the **untreated** unit with the most similar X and use its Y as the missing $Y_i(0)$.

Unit	D	Age	Educ.	Y(1)	Y(0)	Matched to	$\hat{\tau}_i = Y(1) - Y(0)$
A	1	24	10	**9.8**	**8?0**	**#1**	**+1.8**
B	1	31	9	**7.1**	**6?0**	**#3**	**+1.1**
C	1	22	12	**11.5**	**9?9**	**#4**	**+1.6**
D	1	28	11	**8.4**	**7?0**	**#6**	**+1.4**
#1	0	25	10	?	**8.0**		
#2	0	45	16	?	**21.3**		
#3	0	30	9	?	**6.0**		
#4	0	23	12	?	**9.9**		
#5	0	52	13	?	**17.5**		
#6	0	29	11	?	**7.0**		

$$\widehat{\text{ATT}} = \frac{1}{N_1} \sum_{i:\, D_i = 1} \left(Y_i - Y_{j(i)} \right) = \mathbf{+1.48}$$

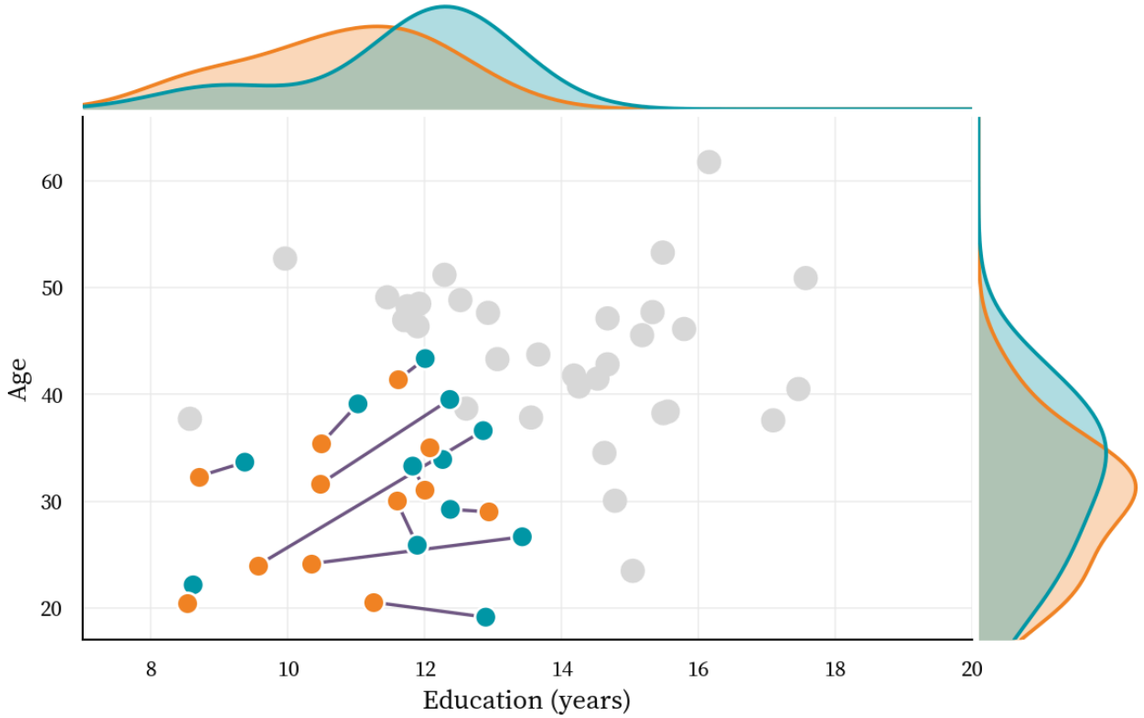
Building intuition: two covariates



1. Two covariates, education and age: each dot is one person.
2. Their distributions (top: education, right: age) **differ**: treated are younger and less educated.
3. So do the means (dotted lines). Comparing all of them is comparing apples and oranges.

Nearest-neighbour matching

Distance between treated i and untreated j (Mahalanobis, scale-free): $d(i,j) = \sqrt{(X_i - X_j)' \, S^{-1} \, (X_i - X_j)}$, with S the covariance matrix of X.

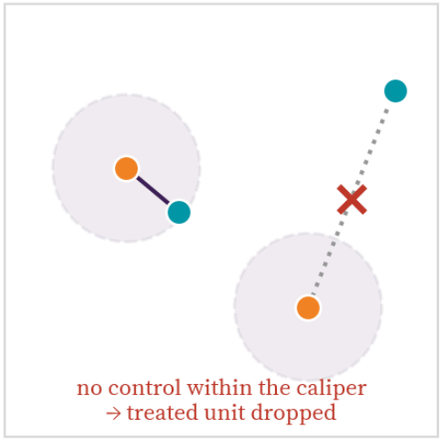


NEAREST-NEIGHBOUR MATCHING

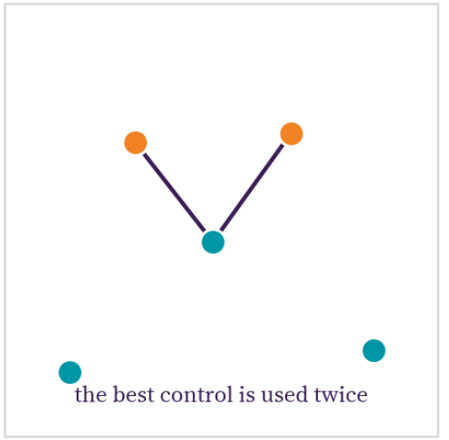
1. Start from all treated and untreated units.
2. Take one treated unit and find its **nearest** untreated neighbour (dashed ellipses: points at the same distance).
3. Repeat for every treated unit: closest pairs first, each untreated used once (no replacement).
4. Drop the untreated units that were not used: the matched groups now **look alike** (top and right).

Choices you have to make

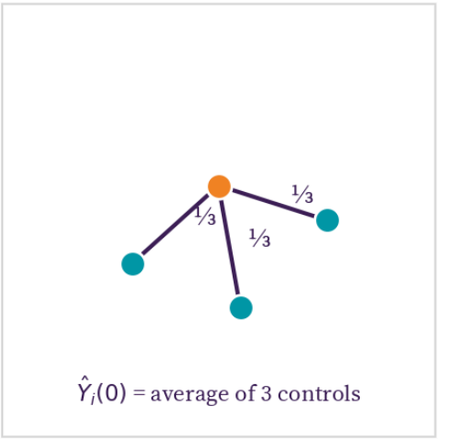
Caliper: a maximum distance



With replacement

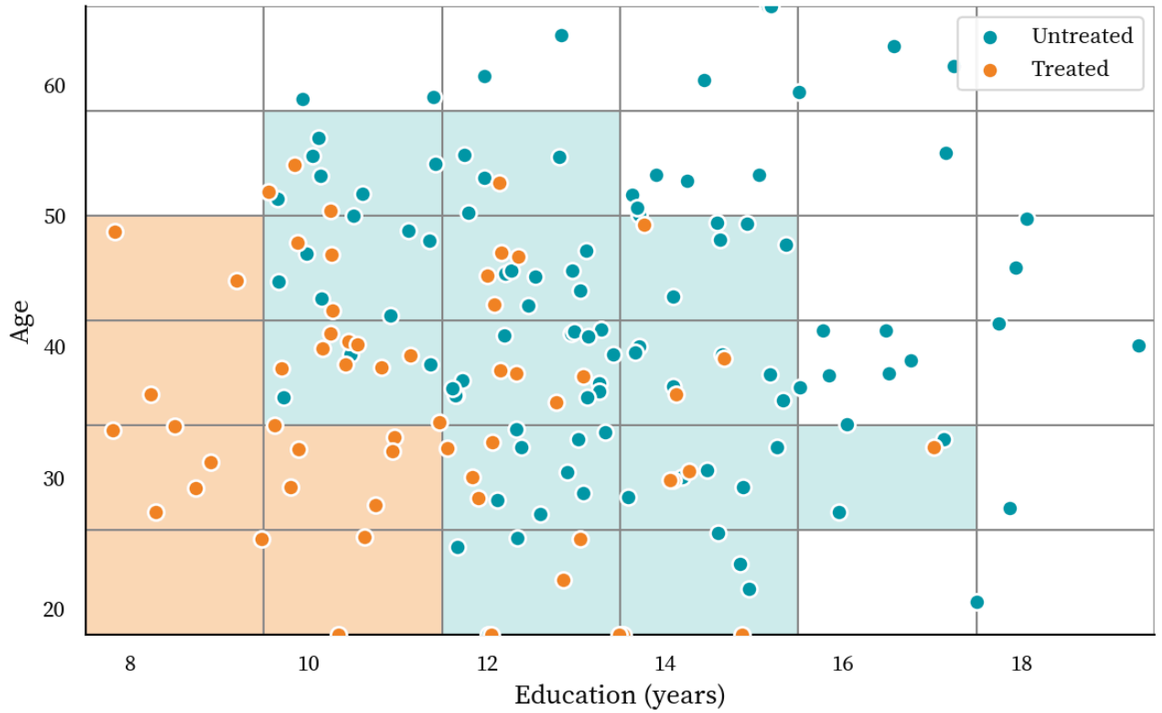


k = 3 neighbours



Choice	Bias	Variance
Tighter caliper	↓ (closer matches)	↑ (fewer treated kept) — and the estimand changes!
With replacement	↓ (always the closest control)	↑ (few controls used many times)
More neighbours k	↑ (the 3rd closest is worse)	↓ (average of more controls)

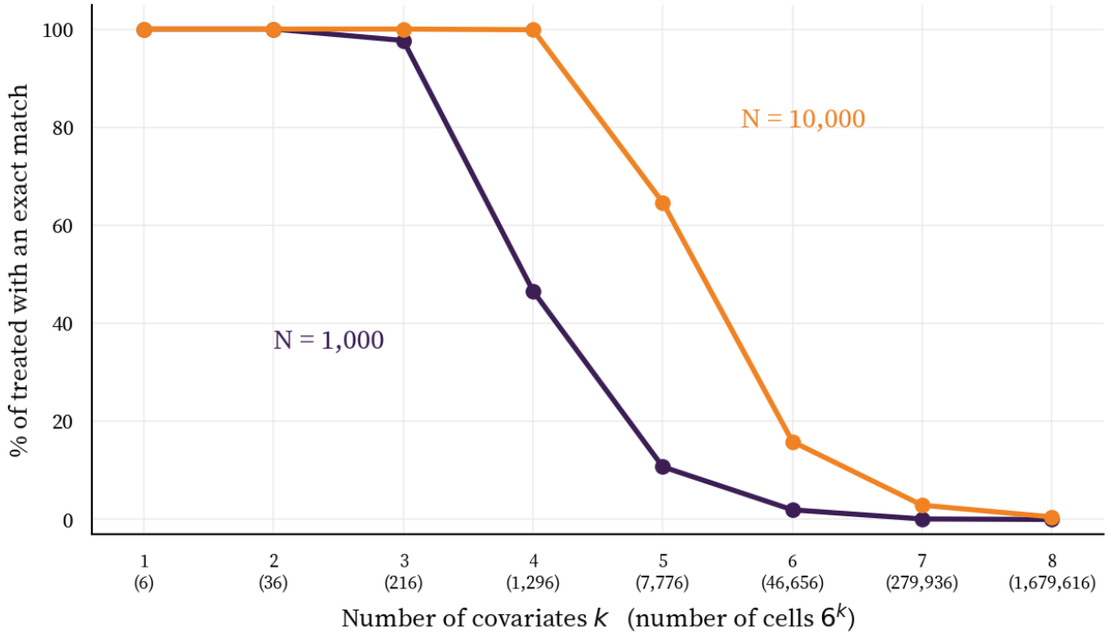
Exact matching



EXACT MATCHING

1. Compare a treated unit only with untreated units that have **exactly the same X**.
2. With continuous X, first make groups: 6 education × 6 age groups = **36 cells**. Same cell = same X.
3. ▯ treated **and** untreated in the cell: exact matches exist.
   ▯ treated but **no** untreated: 16 of 58 treated have no match and are dropped.

Exact matching and the curse of dimensionality



1. Each covariate is cut into **6 groups**, as on the previous slide. A cell is one combination of groups, so with k covariates there are $6 \times 6 \times \ldots \times 6 = \mathbf{6^k}$ **cells** (in brackets under the axis).
2. Simulation with **N = 1,000** people (20% treated): with 3 covariates 98% of the treated find an exact match; with 5 only 11%; with 8, 0%.
3. **N = 10,000**: ten times more data only delays the problem (65% with 5 covariates, 1% with 8). Cells grow ×6 with each covariate, the data do not: the **curse of dimensionality**.

⇒ We need to summarize all of $\boldsymbol{X}$ in **one number**: the propensity score.

3 · The propensity score

The propensity score

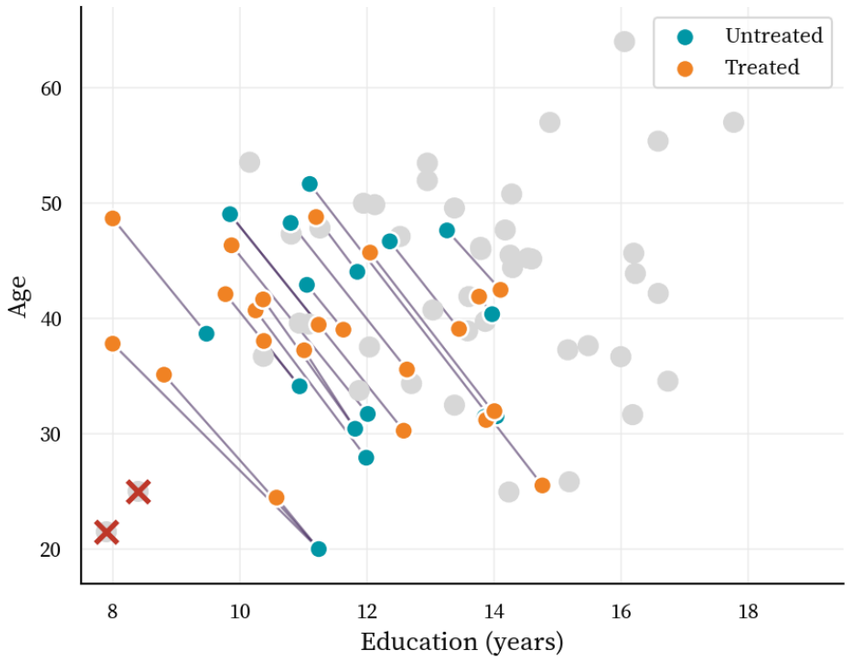
$$p(X) = P(D = 1 \mid X)$$

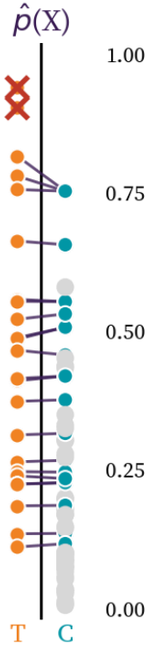
Rosenbaum & Rubin (1983). If the CIA holds given X, it also holds given the scalar $p(X)$:

$$\big(Y(1), Y(0)\big) \perp\!\!\!\perp D \mid X \quad \Longrightarrow \quad \big(Y(1), Y(0)\big) \perp\!\!\!\perp D \mid p(X)$$

- Why? $p(X)$ is a **balancing score**: among units with the same $p(X)$, the distribution of X is the same for treated and untreated, $D \perp\!\!\!\perp X \mid p(X)$.
- So instead of matching on k covariates, we match on **one** number.
- $p(X)$ is unknown: we **estimate** it, usually with a logit or probit $\hat{p}(X_i) = \Lambda(X_i'\hat{\beta})$.

Matching on the propensity score: 2D → 1D → 2D



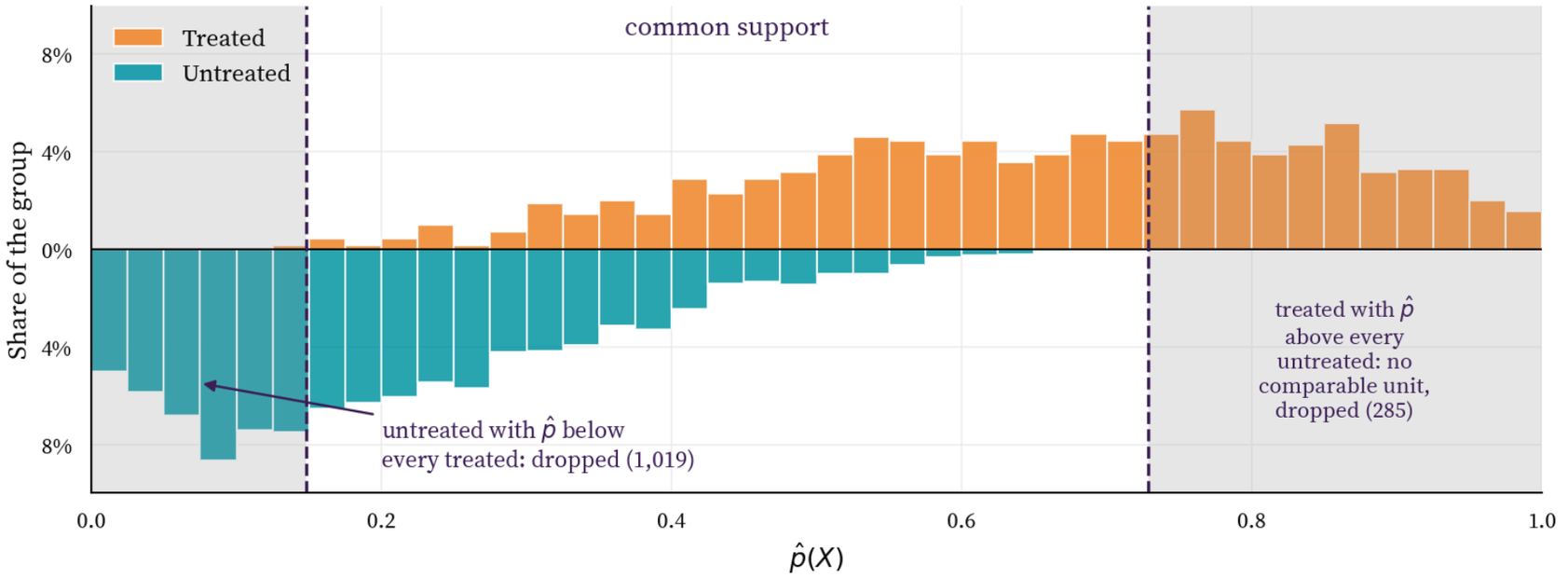


PROPENSITY-SCORE MATCHING

1. Two covariates: treated and untreated differ in X.
2. Estimate $\hat{p}(X)$ with a logit and send every unit to its $\hat{p}$: **one number** per unit (3 clicks).
3. Match each treated unit to the untreated unit with the closest $\hat{p}$ (with replacement, caliper = 0.07).
4. Unused untreated drop out. The 2 treated with $\hat{p}$ far above every untreated have no match within the caliper (×): dropped.
5. Send the matched units back to the 2D space.
6. Pairs share $\hat{p}$, **not X**. But as groups they now look alike: education 11.5 vs 11.8 (before 11.2 vs 13.3), age 38.3 vs 36.4 (before 37.1 vs 42.2).

Then estimate the ATT on the matched sample exactly as before: $\widehat{\mathrm{ATT}} = \frac{1}{N_1} \sum_{i:\, D_i=1} \left(Y_i - Y_{j(i)}\right)$.

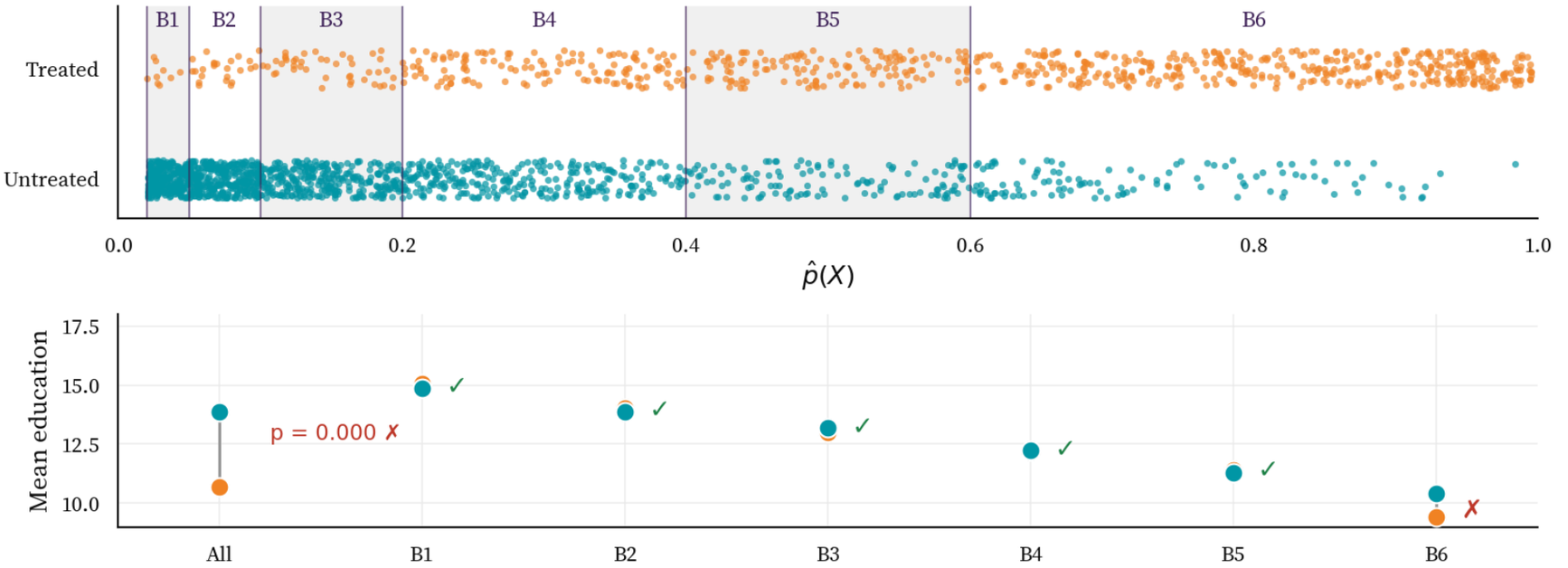
Common support



Stata's `comsup` option keeps only units with $\hat{p}$ in $[\min \hat{p}_{D=1},\ \max \hat{p}_{D=1}]$. Dropping treated units changes **who** the ATT is about: always report how many you lose.

Checking the balancing property: stratify on $\hat{p}(X)$

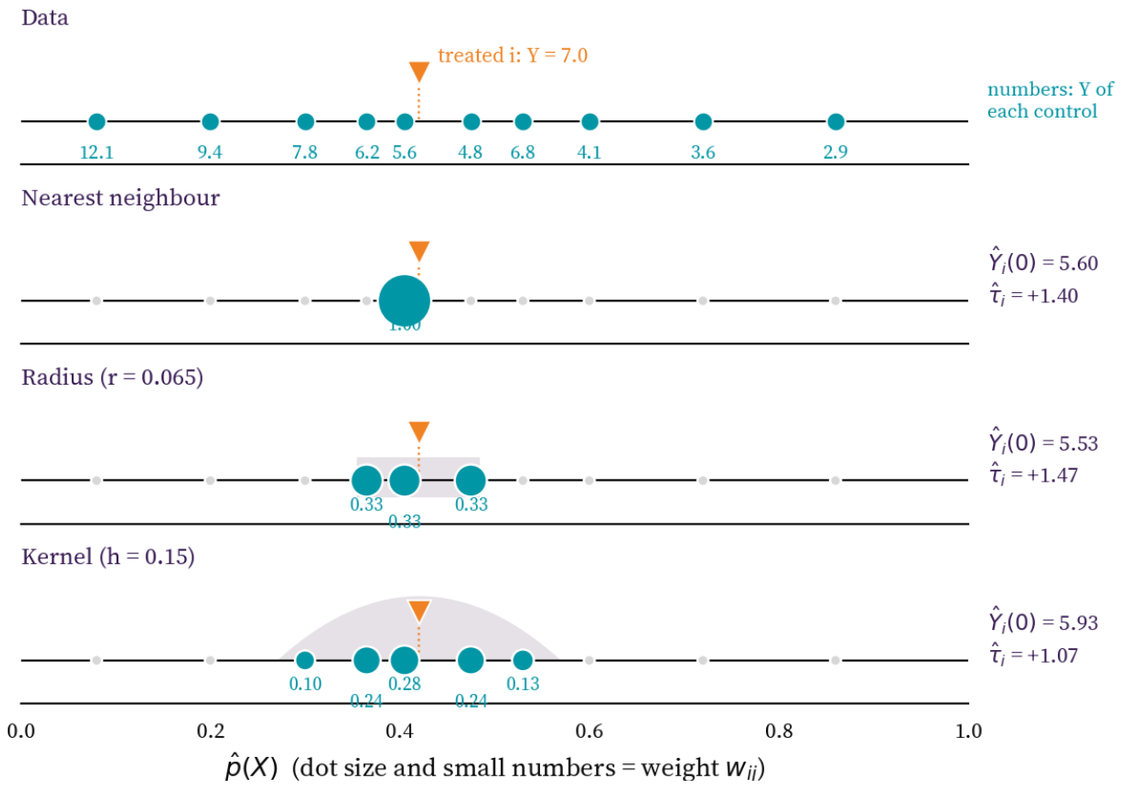
`pscore` (Becker & Ichino, 2002) splits $\hat{p}$ into **blocks** and tests, inside each block, that treated and untreated have the same mean of every X: $D \perp\!\!\!\perp X \mid p(X)$.



If some X fails in some block, Stata says *"The balancing property is not satisfied — try a different specification"*: add **interactions** or **higher powers** to the logit and run again.

From $\hat{p}$ to $\hat{Y}_i(0)$: nearest neighbour, radius or kernel

Once every unit has its $\hat{p}$, we still have to decide **which controls** stand in for each treated unit, and with **which weights**: $\hat{Y}_i(0) = \sum_j w_{ij} Y_j$.

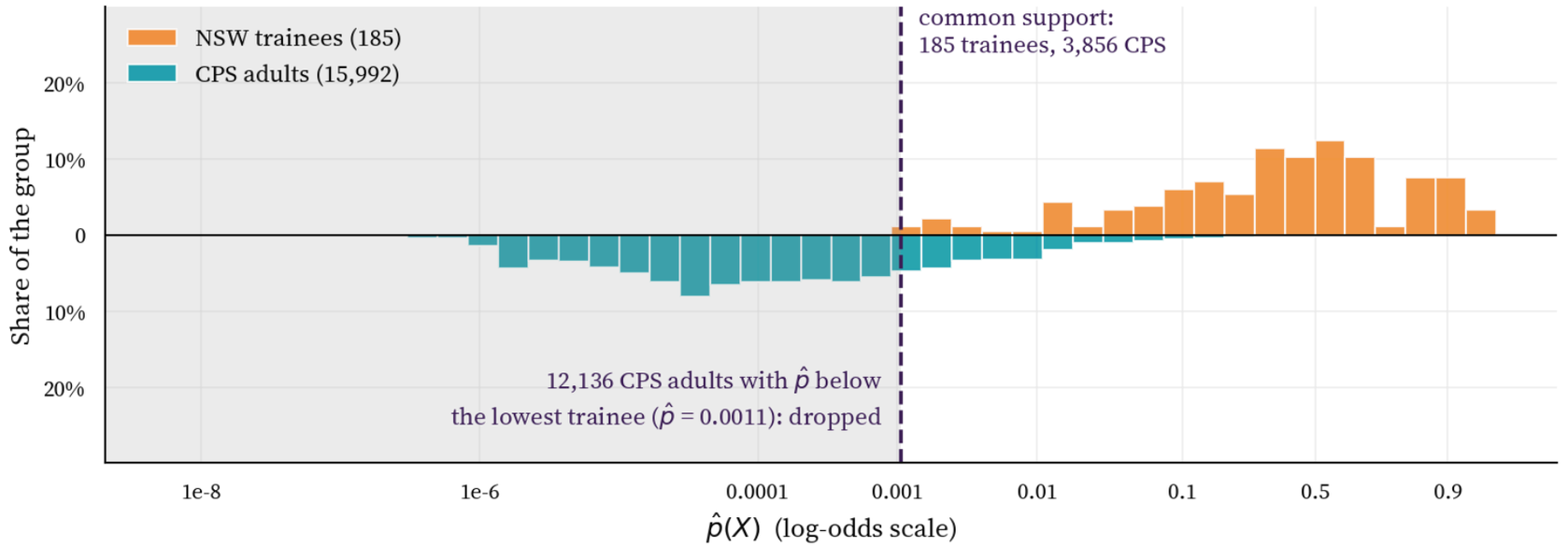


THREE WAYS TO BUILD $\hat{Y}_i(0)$

1. One treated unit i (▼, $\hat{p} = 0.42$) and ten controls, each with its $\hat{p}$ and its Y.
2. **Nearest neighbour**: only the closest control, weight 1 (with k neighbours, 1/k each). What we did so far.
3. **Radius**: every control within a distance r of $\hat{p}_i$, equal weights. More controls, but you have to choose r.
4. **Kernel**: every control, weights falling with the distance (bandwidth h). Uses almost all the data → less variance; no analytical SE ⇒ **bootstrap**.

4 · Back to job training: which matching method wins?

Common support in the NSW data



Most CPS adults have $\hat{p} \approx 0$: nobody like them enrolls in the program. They are **useless** as counterfactuals, and matching ignores them.

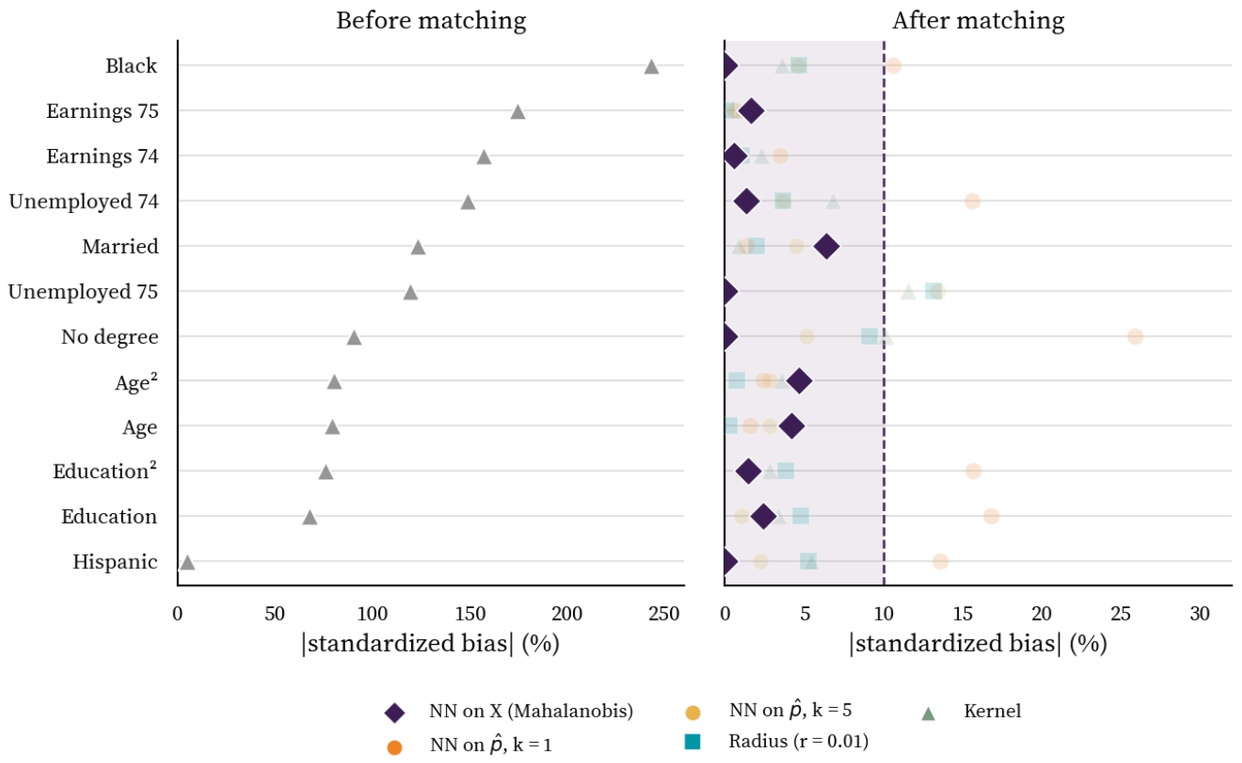
Results: the effect of training with five matching methods

Estimator	ATT (US$)	SE	Participants matched	Controls used	Mean |bias| (%)
Naive difference in means	-8,498	(582)	185	15,992	113.7
OLS + X	1,184	(628)	185	15,992	–
NN on X (Mahalanobis) 🏆	**2,035**	**(948)**	**185**	**118**	**2.1**
NN on $\hat{p}$, k = 1	1,243	(1,115)	178	105	9.1
NN on $\hat{p}$, k = 5	1,859	(884)	178	361	3.4
Radius (r = 0.01)	1,625	(975)	166	3,850	3.9
Kernel (bootstrap SE)	1,740	(840)	178	3,856	4.6

Outcome: earnings in 1978. All methods on the **common support** (185 participants, 3,856 CPS adults). *Participants matched*: those with a comparable control. *Controls used*: CPS adults with positive weight. *Mean |bias|*: average standardized bias of the 12 covariates after matching. SEs from psmatch2 (they ignore that $\hat{p}$ is estimated); kernel SE bootstrapped (50 replications).

- Every matching method turns −$8,498 into a **positive** effect, between $1,243 and $2,035: the sign flip is robust, the size depends on the method.
- NN uses few controls (105–361) → larger SEs. Radius and kernel use (almost) all CPS adults on the common support → smaller SEs.
- Which number do we trust? Not the one we like: the one with the **best balance** (next slide).

Which method balances the covariates best?

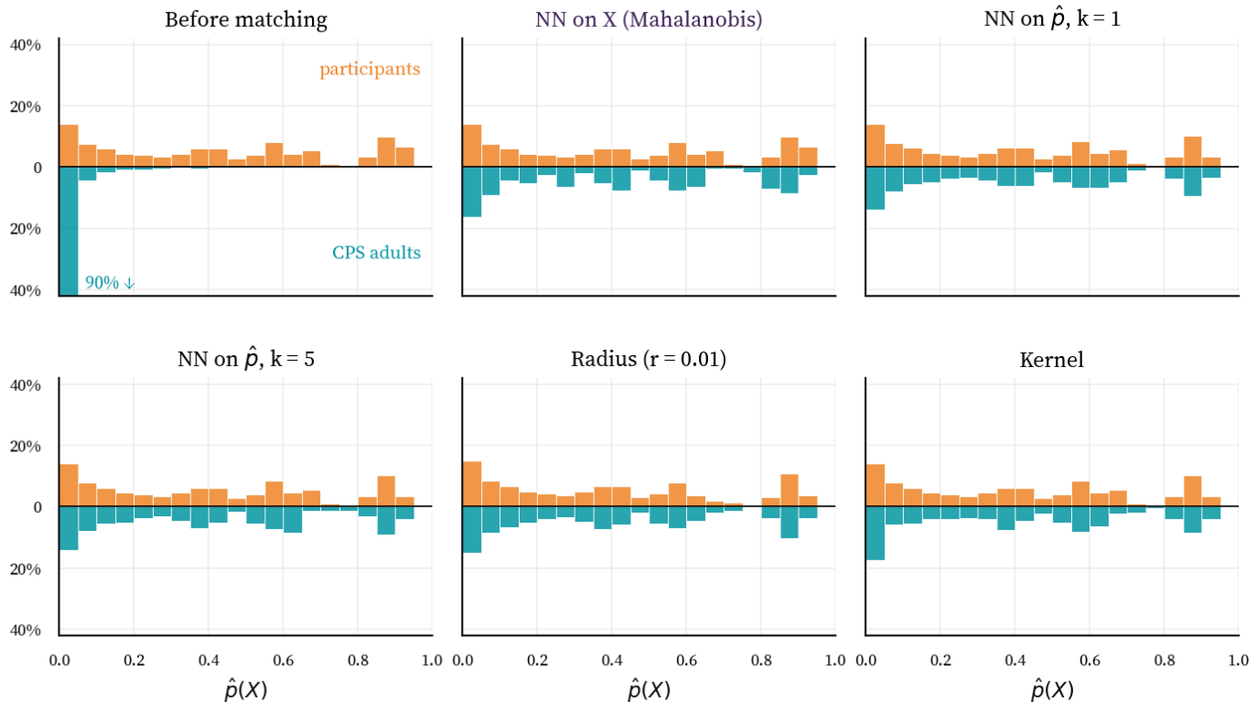


LOVE PLOT: BALANCE BY METHOD

1. **Before matching**: participants and CPS adults differ by up to 243% of a standard deviation (mean 114%).
2. **After matching**, every method removes most of the imbalance (note the different scale).
3. 🏆 **NN on X (Mahalanobis)**: mean |bias| 2.1%, largest 6.4%. Worst: NN on $\hat{p}$, k = 1 (9.1%).

Choose the method by **balance**, decided *before* looking at the ATT, never by the ATT you prefer. Here the winner gives **$2,035**.

The propensity score before and after matching



HOW TO READ IT

1. **Before matching**: $\hat{p}$ of the participants (up) and of the CPS adults on the common support (down). Almost all CPS adults sit at $\hat{p} < 0.05$.
2. **After matching**: each CPS adult weighted by how much it is used. The matched CPS distribution now **mirrors** the participants': the groups are comparable in $\hat{p}$.

Same check as the love plot, but for $\hat{p}$. Even matching on X (Mahalanobis), which never uses $\hat{p}$, balances it. How: each CPS adult enters the histogram with its **matching weight** (how many times, or how much, it is used).

Plot twist: the NSW was an experiment

The NSW participants were **randomly drawn** from the applicants; the 260 applicants randomized out form an **experimental control group**. So for this program we *know* the answer.

185
NSW participants
D = 1

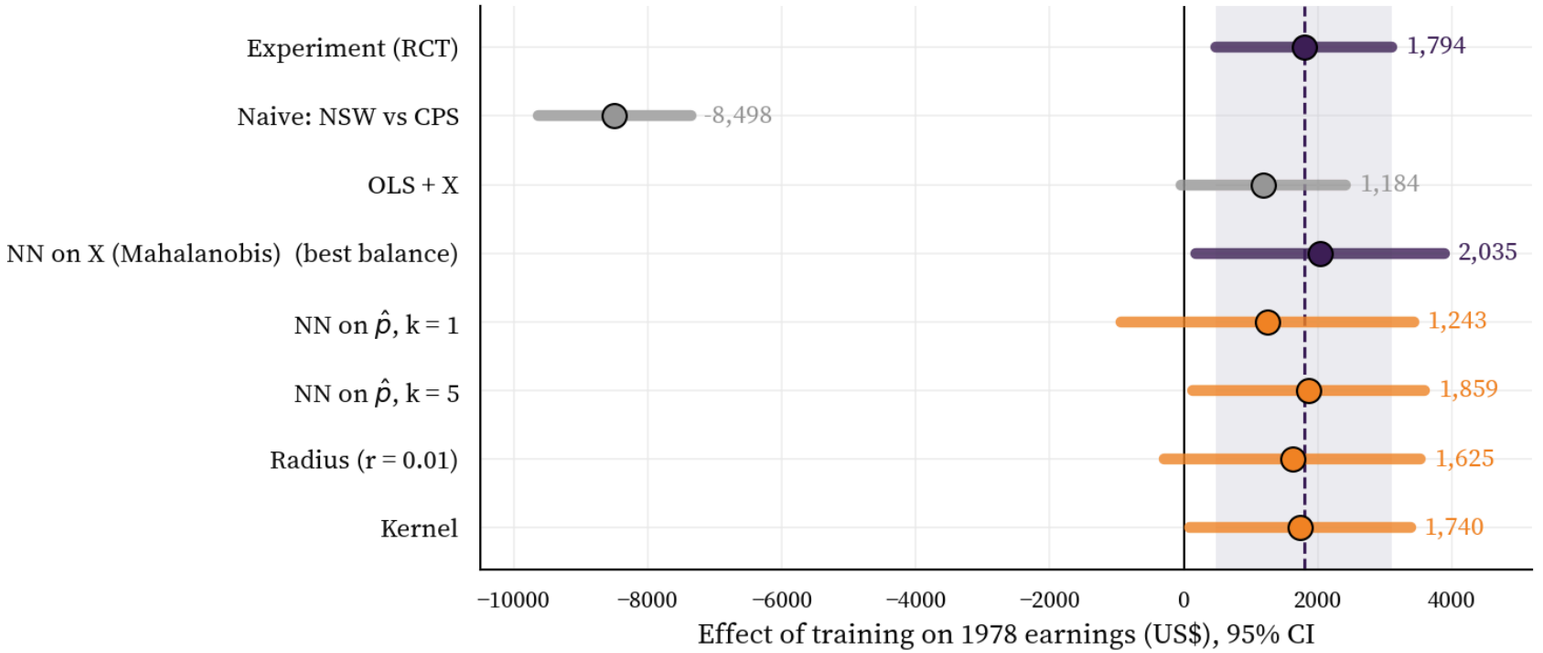
260
NSW randomized controls
the experimental D = 0

15,992
CPS adults
a survey: non-experimental D = 0

Comparison	Effect on 1978 earnings (US$)	SE
Experiment: participants vs randomized controls	**1,794**	(671)
Naive: participants vs CPS adults	−8,498	(582)
OLS + X: participants vs CPS adults	1,184	(628)

LaLonde (1986) did exactly what we did today: throw away the experimental controls, use the CPS, and check whether non-experimental methods recover 1,794. Dehejia & Wahba (1999) showed that **matching** gets close.

Did matching recover the experiment?



Once participants are compared with **comparable** CPS adults, every matching estimate lands inside the experimental 95% CI (shaded), the best-balanced one included. **But** we were lucky to observe the right $\boldsymbol{X}$ (past earnings!): without them the CIA fails and no matching method can help (Smith & Todd, 2005).

Thanks!